Concurrency and Distributed Systems

Week 10 – Part 3: Toxiproxy Failure Injection

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Part 3 Roadmap

1. Tool
2. Experiments
3. Demo

Why Toxiproxy?

- Makes failure reproducible.
- Injects latency, bandwidth limits, resets, and timeouts.
- Lets students compare baseline with controlled damage.

Start Tool

```
docker compose up -d
```

Course note

Java tests do not require Docker. Toxiproxy is for the live failure-injection lab.

Experiment Order

1. Baseline with no toxic.
2. Add latency.
3. Limit bandwidth.
4. Reset or timeout connection.
5. Record throughput, latency, and timeout count.

Live Demo

Run

```
.\run-demo.ps1 04
```